

Deploying AI and ML Models in the Cloud

Artificial intelligence and machine learning are being increasingly used by businesses to enhance productivity. These technologies are also being deployed in the cloud with the help of open source tools and platforms. An important platform in this domain is Hugging Face, which has turned into an indispensable tool for modern AI-driven organisations due to its flexibility and scalability.

Artificial intelligence (AI) and machine learning (ML) are transformative technologies that have revolutionised various industries by enabling machines to mimic human intelligence and learn from data without explicit programming. AI encompasses a wide array of subfields, including natural language processing, computer vision, robotics, and more, each contributing to the development of systems capable of performing tasks that typically require human intelligence. Machine learning, a subset of AI, focuses on algorithms and statistical models that allow computers to identify patterns and make decisions with minimal human intervention. This paradigm shift towards automation and intelligent systems is driving significant innovation across sectors such as healthcare, finance, manufacturing, and more.

The rapid advancements in AI and ML have not only spurred tech growth but have also led to an exponential increase in market size. As organisations recognise the potential of these technologies to optimise processes, enhance customer experiences, and generate actionable insights from vast datasets, investment in AI and ML is soaring. According to recent market analyses, the global AI market is poised to grow at a compound annual growth rate (CAGR) of over 40%, potentially reaching several hundred billion dollars within the next decade. This surge is

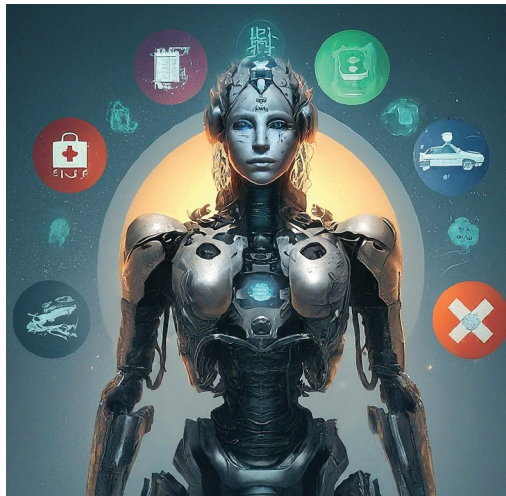


Figure 1: A humanoid robot, representing the future of AI



Figure 2: A cityscape with AI-powered vehicles and drones, showcasing the potential of AI in transportation

attributed to the increasing adoption of AI-driven solutions, such as predictive analytics, autonomous vehicles, and

personalised recommendations, which are becoming integral to business strategies worldwide.

The market dynamics of AI and ML are also shaped by the growing demand for cloud-based services and the integration of AI into Internet of Things (IoT) devices. These trends are facilitating the widespread deployment of AI applications, further expanding the market size. Additionally, as AI and ML technologies continue to evolve, they are expected to drive the emergence of new business models and industries, creating a ripple effect that will influence the global economy at large. Consequently, the AI and ML market is not just expanding in terms of revenue, but also in its impact on the broader technological landscape and societal advancement.

According to Precedence Research (Source: precedenceresearch.com/artificial-intelligence-market/), the global artificial intelligence (AI) market is currently valued at approximately US\$ 638.23 billion and is projected to experience substantial growth over the next decade. By 2034, the market is expected to reach an estimated value of around US\$ 3,680.47

billion, driven by a robust compound annual growth rate (CAGR) of 19.1% from 2024 to 2034. North America continues to be a